Customer Shopping Behavior Analysis

End-to-End Data Analytics Project

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Tools Used: Python | PostgreSQL | Power BI

Dataset Source: Kaggle

99,457	12	3	■251.51M
Total Records	Columns	Tools Used	Total Revenue

1. Project Overview

This project presents a complete end-to-end data analytics solution for analyzing customer shopping behavior across multiple shopping malls in Istanbul, Turkey. The analysis covers data cleaning in Python, structured querying in PostgreSQL, and interactive visualization through a Power BI dashboard.

Business Problem Statement

A retail shopping company wants to understand customer purchasing behavior across malls, product categories, age groups, and genders in order to improve sales performance and optimize marketing strategies.

Core Business Question: "How can the company leverage consumer shopping data to identify trends, improve customer engagement, and optimize marketing and product strategies?"

Project Workflow

Step 1	Step 2	Step 3
Python (Data Cleaning)	**PostgreSQL (SQL Analysis)**	**Power BI (Dashboard)**
Clean & prepare data	Answer 10 business questions with SQL	Interactive visual dashboard

Dataset Information

Dataset Name	Customer Shopping Dataset – Retail Sales Data
Source	Kaggle
Location	Istanbul, Turkey
Currency	Turkish Lira (TL)
Total Rows	99,457
Original Columns	10
Final Columns	12 (after adding total_amount & age_group)
File Format	CSV

2. Step 1 – Data Cleaning (Python)

Libraries Used

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
print("Libraries loaded successfully!")
```

Output: Libraries loaded successfully!

Step 1 – Load Dataset

```
df = pd.read_csv(r"customer_shopping_data.csv")
print("Dataset loaded!")
print(df.shape)
```

Output: Dataset loaded! Shape: (99457, 10)

Step 2 – View First 5 Rows (df.head())

invoice_no	customer_id	gender	age	category	quantity	price	payment_method	invoice_date	shopping_mall
I138884	C241288	Female	28	Clothing	5	1500.40	Credit Card	05-08-2022	Kanyon
I317333	C111565	Male	21	Shoes	3	1800.51	Debit Card	12-12-2021	Forum Istanbul
I127801	C266599	Male	20	Clothing	1	300.08	Cash	09-11-2021	Metrocity
I173702	C988172	Female	66	Shoes	5	3000.85	Credit Card	16-05-2021	Metropol AVM
I337046	C189076	Female	53	Books	4	60.60	Cash	24-10-2021	Kanyon

Step 3 – Dataset Info (df.info())

```
RangeIndex: 99457 entries, 0 to 99456
Data columns (total 10 columns):
# Column Non-Null Count Dtype
0 invoice_no 99457 non-null str
1 customer_id 99457 non-null str
2 gender 99457 non-null str
3 age 99457 non-null int64
4 category 99457 non-null str
5 quantity 99457 non-null int64
6 price 99457 non-null float64
7 payment_method 99457 non-null str
8 invoice_date 99457 non-null str
9 shopping_mall 99457 non-null str
Memory usage: 7.6 MB
```

Step 4 – Statistical Summary (df.describe())

Statistic	age	quantity	price
count	99457	99457	99457
mean	43.43	3.00	689.26
std	14.99	1.41	941.18
min	18.00	1.00	5.23
25%	30.00	2.00	45.45

50% (median)	43.00	3.00	203.30
75%	56.00	4.00	1200.32
max	69.00	5.00	5250.00

Step 5 – Check Missing Values (df.isnull().sum())

```
invoice_no 0
customer_id 0
gender 0
age 0
category 0
quantity 0
price 0
payment_method 0
invoice_date 0
shopping_mall 0
dtype: int64
```

Result: No missing values found in any column. Dataset is complete.

Step 6 – Check & Remove Duplicates

```
print("duplicate rows:", df.duplicated().sum())
df = df.drop_duplicates()
print("clean dataset shapes:", df.shape)
```

Output: duplicate rows: 0 | clean dataset shapes: (99457, 11)

Step 7 – Standardize Column Names

```
df.columns = df.columns.str.strip().str.lower().str.replace(' ','_')
```

Output: columns names renamed

Step 8 – Create total_amount Column

```
df['total_amount'] = (df['quantity'] * df['price']).round(2)
print("Total amount column created")
df.head()
```

Output: Total amount column created (quantity × price, rounded to 2 decimal places)

Step 9 – Category Distribution

```
Clothing 34487
Cosmetics 15097
Food & Beverage 14776
Toys 10087
Shoes 10034
Souvenir 4999
Technology 4996
Books 4981
```

Step 10 – Gender Distribution

```
Female 59482
Male 39975
```

Step 11 – Create Age Group Column

```
bins = [0, 19, 25, 40, 58, 80]
labels = ['Young','Young Adult','Adult','Middle Aged','Senior']
```

```
df['age_group'] = pd.cut(df['age'], bins=bins, labels=labels)
```

Age Group	Age Range
Young	0 – 19
Young Adult	20 – 25
Adult	26 – 40
Middle Aged	41 – 58
Senior	59 – 80

Step 12 – Export Clean Dataset

```
df.to_csv("customer_shopping_data_clean.csv", index=False)
```

Output: Clean data exported successfully! Final shape: (99457, 12)

Data Cleaning Summary

Check Performed	Result	Action Taken
Missing Values	0 missing values found	No action required
Duplicate Rows	0 duplicates found	No action required
Column Names	Standardized to lowercase	str.lower() + str.replace()
Date Column	Converted to datetime	pd.to_datetime()
Total Amount	New column created	quantity x price (rounded 2dp)
Age Group	New column created	pd.cut() with 5 categories

3. Step 2 – SQL Analysis (PostgreSQL)

The cleaned dataset was imported into PostgreSQL as a table named 'customers'. Ten business questions were answered using SQL queries covering basic aggregations, subqueries, RANK(), ROW_NUMBER() window functions, and CTEs.

SQL Concept	Used In
GROUP BY & ORDER BY	Q1, Q2, Q3, Q4, Q5
ROUND & Aggregate Functions	All Queries
Subqueries	Q6, Q7
RANK() Window Function	Q6, Q8
ROW_NUMBER() Window Function	Q10
CTE (WITH clause)	Q9
Running Total (SUM OVER)	Q9
PARTITION BY	Q6, Q10

Q1. Which product category contributes the most to overall revenue?

SQL Query:

```
SELECT category,
ROUND(SUM(total_amount)::numeric, 2) AS total_revenue
FROM customers
GROUP BY category
ORDER BY total_revenue DESC
LIMIT 1;
```

Result:

category	total_revenue (TL)
Clothing	113,996,791.04

Finding: Clothing is the top revenue-generating category with total revenue of TL 113,996,791.04.

Q2. Which shopping mall is the top performing location in terms of total sales?

SQL Query:

```
SELECT shopping_mall,
ROUND(SUM(total_amount)::numeric, 2) AS sales
FROM customers
GROUP BY shopping_mall
ORDER BY sales DESC
LIMIT 1;
```

Result:

shopping_mall	sales (TL)
Mall of Istanbul	50,872,481.68

Finding: Mall of Istanbul is the highest-grossing location with total sales of TL 50,872,481.68.

Q3. What is the most preferred payment method among customers?

SQL Query:

```
SELECT payment_method,
COUNT(*) AS total_transactions
FROM customers
GROUP BY payment_method
ORDER BY total_transactions DESC;
```

Result:

payment_method	total_transactions
Cash	44,447
Credit Card	34,931
Debit Card	20,079

Finding: Cash is the most preferred payment method with 44,447 transactions, followed by Credit Card (34,931) and Debit Card (20,079).

Q4. Which gender segment drives higher purchasing power and what is their average spending?

SQL Query:

```
SELECT gender,
ROUND(SUM(total_amount)::numeric, 2) AS total_purchase,
ROUND(AVG(total_amount)::numeric, 2) AS average_spending
FROM customers
GROUP BY gender
ORDER BY total_purchase DESC;
```

Result:

gender	total_purchase (TL)	average_spending (TL)
Female	150,207,136.02	2,525.25
Male	101,298,658.23	2,534.05

Finding: Female customers drive higher total purchasing power (TL 150.2M) however male customers have a slightly higher average spending per transaction (TL 2,534.05 vs TL 2,525.25).

Q5. Which age group represents the most valuable customer segment?

SQL Query:

```
SELECT age_group,
ROUND(SUM(total_amount)::numeric, 2) AS total_spending,
ROUND(AVG(total_amount)::numeric, 2) AS average_spending
FROM customers
GROUP BY age_group
ORDER BY total_spending DESC, average_spending;
```

Result:

age_group	total_spending (TL)	avg_spending (TL)
Middle Aged	87,245,011.86	2,547.30
Adult	73,299,544.08	2,533.60
Senior	52,842,966.96	2,526.32
Young Adult	29,067,212.46	2,510.34
Young	9,051,058.89	2,394.46

Finding: Middle Aged customers (41-58 years) represent the most valuable segment with TL 87.2M in total spending and the highest average spending of TL 2,547.30.

Q6. What is the top purchasing category for each gender?

SQL Query:

```
SELECT gender, category, total_purchases
FROM (
SELECT gender, category, COUNT(*) AS total_purchases,
RANK() OVER (PARTITION BY gender ORDER BY COUNT(*) DESC) AS rank
FROM customers GROUP BY gender, category
) ranked WHERE rank = 1;
```

Result:

gender	category	total_purchases
Female	Clothing	20,652
Male	Clothing	13,835

Finding: Clothing is the most purchased category for both Female (20,652 purchases) and Male (13,835 purchases) customers.

Q7. Which shopping mall attracts the highest number of young adult customers?

SQL Query:

```
SELECT shopping_mall, number_of_customers
FROM (
SELECT shopping_mall, age_group,
COUNT(*) AS number_of_customers
FROM customers GROUP BY shopping_mall, age_group
) AS subquery
WHERE age_group = 'Young Adult'
```

```
ORDER BY number_of_customers DESC;
```

Result:

shopping_mall	number_of_customers
Mall of Istanbul	2,382
Kanyon	2,337
Metrocity	1,700
Metropol AVM	1,225
Istinye Park	1,122

Finding: Mall of Istanbul attracts the most Young Adult customers with 2,382 visits, closely followed by Kanyon (2,337).

Q8. How do shopping malls rank against each other in terms of total revenue?

SQL Query:

```
SELECT shopping_mall,
ROUND(SUM(total_amount)::numeric, 2) AS total_revenue,
RANK() OVER (ORDER BY SUM(total_amount) DESC) AS revenue_rank
FROM customers GROUP BY shopping_mall;
```

Result:

shopping_mall	total_revenue (TL)	revenue_rank
Mall of Istanbul	50,872,481.68	1
Kanyon	50,554,231.10	2
Metrocity	37,302,787.33	3
Metropol AVM	25,379,913.19	4
Istinye Park	24,618,827.68	5
Zorlu Center	12,901,053.82	6
Cevahir AVM	12,645,138.20	7
Viaport Outlet	12,521,339.72	8
Emaar Square Mall	12,406,100.29	9
Forum Istanbul	12,303,921.24	10

Finding: Mall of Istanbul ranks 1st (TL 50.87M) with Kanyon very close in 2nd place (TL 50.55M). The top 3 malls collectively generate over 55% of total revenue.

Q9. What is the running total of revenue generated day by day?

SQL Query:

```
WITH daily_revenue AS (
SELECT invoice_date,
ROUND(SUM(total_amount)::numeric, 2) AS total_revenue
FROM customers GROUP BY invoice_date
)
SELECT invoice_date, total_revenue,
ROUND(SUM(total_revenue) OVER
(ORDER BY invoice_date)::numeric, 2) AS running_total
FROM daily_revenue ORDER BY invoice_date;
```

Result:

invoice_date	daily_revenue (TL)	running_total (TL)
01-01-2021	271,192.66	271,192.66
01-01-2022	179,147.49	450,340.15
01-01-2023	426,016.12	876,356.27
01-02-2021	265,837.38	1,142,193.65
01-02-2022	336,410.42	1,478,604.07
... (continues)	...	...

Finding: Revenue accumulates consistently across all dates. A CTE calculated daily totals and a SUM() window function computed the cumulative running total, revealing steady business growth over time.

Q10. Who is the top spending customer in each age group?

SQL Query:

```
SELECT customer_id, age_group, total_spent
FROM (
SELECT customer_id, age_group,
ROUND(SUM(total_amount)::numeric, 2) AS total_spent,
ROW_NUMBER() OVER
(PARTITION BY age_group ORDER BY SUM(total_amount) DESC) AS rank
FROM customers GROUP BY customer_id, age_group
) ranked WHERE rank = 1 ORDER BY total_spent DESC;
```

Result:

customer_id	age_group	total_spent (TL)
C600068	Adult	26,250.00
C354711	Middle Aged	26,250.00
C984334	Senior	26,250.00
C133108	Young	26,250.00
C233284	Young Adult	26,250.00

Finding: ROW_NUMBER() was used instead of RANK() to return exactly one top customer per age group even when multiple customers share the same total spent amount of TL 26,250.00.

4. Step 3 – Power BI Dashboard

The cleaned PostgreSQL data was connected to Power BI Desktop to build an interactive dashboard. DAX measures were created and visualizations were designed to answer all 10 business questions visually.

DAX Measures Created

Measure Name	DAX Formula
Total Revenue	SUM(customers[total_amount])
Average Spending	AVERAGE(customers[total_amount])
Total Transactions	COUNT(customers[invoice_no])
Total Customers	DISTINCTCOUNT(customers[customer_id])

Dashboard KPI Cards

KPI	Value
Average Spending	TL 2.53K
Total Customers	99K
Total Revenue	TL 251.51M
Total Transactions	99K

Dashboard Visuals

Visual	Chart Type	Fields Used
Total Revenue by Gender	Donut Chart	gender, Total Revenue
Revenue by Age Group	Clustered Column Chart	age_group, Total Revenue
Total Transactions by Category	Clustered Column Chart	category, Total Transactions
Average Spending by Age Group	Clustered Bar Chart	age_group, Average Spending
Revenue by Shopping Mall	Clustered Bar Chart	shopping_mall, Total Revenue
Revenue Trend Over Time	Line Chart	invoice_date, Total Revenue

Interactive Slicers

Slicer	Style	Purpose
Filter by Gender	Button (Male / Female)	Filter all visuals by gender
Payment Method	List (Cash/Credit/Debit)	Filter by payment type
Filter by Category	List	Filter by product category
Shopping Mall	List	Filter by specific mall

5. Key Insights & Recommendations

1. Clothing Dominates Revenue

Clothing alone generated TL 113.99M which is the highest among all categories. The business should prioritize clothing inventory, promotions, and seasonal campaigns to maintain this strong performance.

2. Mall of Istanbul is the Top Performer

Mall of Istanbul leads all locations with TL 50.87M in revenue and also attracts the most Young Adult customers (2,382). Investing further in this location with exclusive offers and events will drive even higher footfall.

3. Female Customers Drive Overall Revenue

Female customers account for TL 150.2M (59.72%) of total revenue with 59,482 customers vs 39,975 males. Targeted marketing campaigns for female shoppers — especially in Clothing and Cosmetics — will yield high returns.

4. Middle Aged Segment is Most Valuable

Middle Aged customers (41-58 years) generated TL 87.24M with the highest average spending of TL 2,547.30. Loyalty programs and premium product offerings tailored to this age group can significantly increase customer lifetime value.

5. Cash Payments Dominate — Go Digital

44,447 transactions were cash-based (44.7% of all transactions). This presents a major opportunity to introduce digital payment incentives such as cashback on Credit Card and Debit Card usage to modernize payment behavior.

6. Top 2 Malls Generate 40%+ of Revenue

Mall of Istanbul and Kanyon together generate over TL 101M — more than 40% of total revenue. Other malls like Zorlu Center and Viaport Outlet generate significantly less and may benefit from targeted promotions and partnerships.

7. Young Customers Have Lowest Spending

The Young age group (under 19) has the lowest total spending (TL 9.05M) and average spending (TL 2,394.46). Introducing student discounts, loyalty cards, and affordable product bundles can attract and retain this future customer segment.

6. Project Files & GitHub Structure

File	Description	Tool
01_data_cleaning.ipynb	Python data cleaning notebook	Jupyter Notebook
02_sql_analysis.sql	10 SQL business queries	PostgreSQL / pgAdmin
03_dashboard.pbix	Interactive Power BI dashboard	Power BI Desktop
customer_shopping_data.csv	Raw original dataset	Kaggle
customer_shopping_data_clean.csv	Cleaned final dataset	Python output
README.md	Project documentation	GitHub